Business Information Systems

Patient Treatment in Brazilian HR Report

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1 Introduction

1.1 Project Objectives

The primary objective of this project is to analyze the end-to-end patient treatment process in a high-volume Brazilian Emergency Room using Process Mining techniques. By applying the Knowledge Uplift Trail (KUT) methodology, this study aims to discover the current and the optimal process, identify structural inefficiencies, and assess process conformance. Ultimately, the goal is to propose data-driven improvements that enhance the efficiency and quality of patient care.

1.2 Dataset Overview

The analysis is based on an anonymized event log provided as an XES-compliant CSV file. The dataset captures the historical execution of the emergency room processes, containing over 3,000 events distributed across 443 unique patient cases. Each row represents a single event nested within a specific case.

The dataset contains several standard process mining attributes alongside domain-specific clinical data. The key attributes utilized in this analysis include:

- `case:concept:name`: The unique identifier representing a single patient's case.
- `concept:name`: The specific activity performed during the process (e.g., *Atendimento*, *Triagem*, *Consulta*, *Exames Laboratoriais*, *Alta* etc.).
- `time:timestamp`: The precise date and time when the event occurred, essential for control-flow discovery and performance analysis.
- `Doena (Disease)`: Clinical attributes describing the patient's diagnosis (e.g., Influenza or Chest Pain).
- `Retorno (Discharge)`: The patient's discharge status and expected return timeframe.
- `Item`: Details regarding specific materials or medications prescribed during the treatment.
- `Mdico Responsvel (Responsible Doctor)`: The specific resource (physician) executing the task.
- `outlier_label`: A pre-computed conformance flag that categorizes cases as *inliers* or *outliers* based on their overall duration.

2 Organizational Goals

The organizational goals are articulated at three levels, from general strategic directions down to specific tactical actions. All objectives are grounded in the findings of the analytical steps carried out in this project.

Strategic goals: Improve the efficiency and quality of patient care procedures in the Pronto Atendimento, reducing the average Length of Stay and ensuring regulatory compliance.

Operational objectives:

- Reduce process inefficiency and identify bottlenecks in the patient journey.
- Ensure that disease-specific patient journeys are managed through appropriate and standardized clinical pathways.
- Optimize resource allocation across physicians.

Tactical actions:

- Analyse case flows using process mining techniques to establish a reference protocol.
- Real-time monitoring to continuously track case flows and detect delays or anomalies in the process execution.
- Automate simple and repetitive activities such as patient registration and triage for low-complexity cases, freeing clinical resources for more critical tasks.
- Enforce triage compliance for all incoming patients to prevent the protocol bypasses.
- Define specific treatment pathways for the most complex and impactful disease that deviate significantly from the previously discovered standard protocol.
- Organise periodic staff meetings to allow physicians to review workload distribution and collectively define standardised disease-specific care protocols.

3 Knowledge Uplift Trail

3.1 Exploratory Analysis

Step	**Input**	**Acquired Knowledge**	**Output**
Variant analysis	Event log	Distribution and coverage of process variants; power-law fit	Variant frequency distribution
Rework analysis	Event log	Frequency of repeated activities per case	Rework counts per activity
Comparative mining	Event log	Correlation between case size and case duration	Case-level KPI table
Process discovery	Event log	Overall process structure and activity flow	DFG + global Petri net

Table 1: KUT – Exploratory Analysis

3.2 Outlier-Inlier Analysis

Step	**Input**	**Acquired Knowledge**	**Output**
Segmentation	Event log	Segmentation of Outlier / Inlier	Outlier / inlier segments
KPI comparison	Outlier / inlier segments	Distribution and significance of KPI differences across segments	KPI table and significant difference per segment
Process discovery	Inlier log (power-law cutoff)	Best-fitting process model for standard cases	Inlier reference Petri net
Conformance check	Outlier log + inlier model	Type and frequency of deviations from the standard model	Alignment deviation chart

Table 2: KUT – Outlier-Inlier Analysis

3.3 Discharged-Readmitted Analysis

Step	**Input**	**Acquired Knowledge**	**Output**
Segmentation	Event log	Segmentation of Discharged / Readmitted	Discharged / Readmitted segments
KPI comparison	Discharged / Readmitted segments	Distribution and significance of KPI differences across segments	KPI table and significant difference per segment
Conformance check	Discharged / Readmitted segments + inlier model	Fitness and problematic transitions per segment	Fitness scores; problematic transition comparison

Table 3: KUT – Discharged-Readmitted Analysis

3.4 Disease Analysis

Step	**Input**	**Acquired Knowledge**	**Output**
Segmentation	Event log	Segmentation of diseases	Diseases segments
KPI comparison	Diseases segments	Distribution and significance of KPI differences across diseases	KPI table and significant difference per segment
Pairwise comparison	Diseases segments	Identification of which specific diseases differ significantly from the others	Pairwise test results
Readmission correlation	Disease + readmission status	Association between clinical diagnosis and readmission rate	Fisher's exact test results
Process discovery	Statistical different diseases	Disease-specific process structure and complexity	Disease-specific Petri nets
Conformance check	Diseases logs + inlier model	Fitness of each diseases segment against the standard inlier model	Fitness per disease

Table 4: KUT – Disease Analysis

3.5 Resource Analysis

Step	**Input**	**Acquired Knowledge**	**Output**
Workload distribution	Event log	Calculate distribution of cases and workload across physicians	Workload index per physician
Overload identification	Workload indexes	Identification of abnormal workload concentration	List of overburdened physicians

Table 5: KUT – Resource Analysis

4 Project Results

The project code has been uploaded into a GitHub repository [1].

4.1 Data Preprocessing

After loading the dataset, an initial inspection was performed to assess its structure and quality. Several issues were identified and addressed sequentially.

First, duplicate rows and events with missing critical attributes (case identifier, activity name, or timestamp) were removed, as these fields are mandatory for any process mining analysis. Then, empty numeric attributes were instead imputed with zero.

```
# Drop duplicates and rows with missing critical values
clean_df = df.drop_duplicates()
clean_df = clean_df.dropna(
    subset=["case:concept:name", "concept:name", "time:timestamp"])

# Fill missing values in "Quantidade" and "Tempo Processo" with 0
clean_df["Quantidade"] = clean_df["Quantidade"].fillna(0)
clean_df["Tempo Processo"] = clean_df["Tempo Processo"].fillna(0)
```

Cases with zero or negative duration were then excluded: without a valid time interval, no performance metric can be computed and the case is considered corrupted.

```
# Sort the dataframe by case and timestamp
clean_df = clean_df.sort_values(
    by=["case:concept:name", "time:timestamp"]
).reset_index(drop=True)

# Filter out cases with non-positive durations
case_durations = clean_df.groupby("case:concept:name")[("time:
                                    timestamp")].agg(["min", "max"])
case_durations["duration_seconds"] = (
    case_durations["max"] - case_durations["min"]
).dt.total_seconds()

valid_cases = case_durations[case_durations["duration_seconds"] > 0].
                                    index
clean_df = clean_df[clean_df["case:concept:name"].isin(valid_cases)]
```

Incomplete traces were filtered by enforcing the expected process boundaries: every valid patient journey must start with *Atendimento* (admission) and end with *Alta* (discharge). Cases not satisfying this constraint were deemed partially recorded and discarded.

```
# Keep only cases with expected start and end activities
clean_df = pm4py.filter_start_activities(clean_df, ["Atendimento"])
clean_df = pm4py.filter_end_activities(clean_df, ["Alta"])
```

Finally, seven attributes irrelevant to the analytical goals were dropped to reduce data volume, and the remaining Portuguese column names were translated to English for consistency with the downstream analysis.

```
# Select useless columns for the event log
useless_columns = ["N vel de Urg ncia", "Idade", "Sexo", "CID",
                   "Setor", "Multipassante", "Conv nio"]
clean_df = clean_df.drop(columns=useless_columns)

# Renaming columns to English and more descriptive names
clean_df = clean_df.rename(columns={
    "M dico Respons vel": "responsible_physician",
    "Doen a": "disease",  "Item": "item",
    "Tempo Processo": "process_time",
    "Quantidade": "quantity",
    "Retorno": "readmission_status"})
```

The resulting preprocessed log contains only well-formed, complete traces with the attributes required for the subsequent segmentation, discovery, and conformance analyses.

4.2 Exploratory Analysis

This section provides an initial characterisation of the event log through variant distribution analysis, case-level KPI inspection, and global process discovery. The goal is to understand the structural properties of the process before applying more focused segmentation techniques.

Variant Distribution

The preprocessed log contains 436 cases distributed across 103 unique process variants. The distribution is heavily skewed: the top 3 variants already account for the large majority of traces, while the remaining variants form a very long tail, most occurring only once. As shown in Figure 1, the frequency drops sharply after the first few ranks, a pattern consistent with a highly variable clinical process where patients follow personalised paths rather than a rigid protocol.

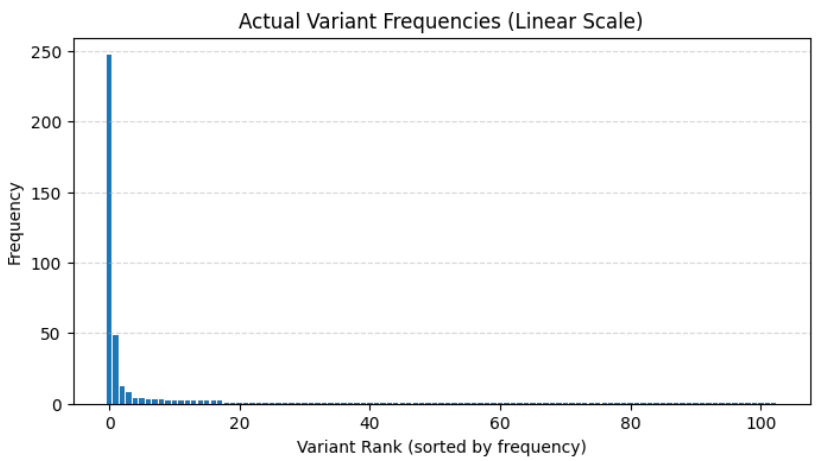


Figure 1: Variant frequency distribution (linear scale), sorted by rank.

Table 6 summarises the three most frequent variants, which cover approximately 69% of cases, and their KPIs.

The simplest variant (the top one) has a median stay of only 57 minutes, while adding a single diagnostic step nearly doubles it, identifying imaging and laboratory exams as structural bottlenecks even in straightforward cases.

To characterise the distributional form, the variant frequencies were fitted against both a power-law and a lognormal model using the `powerlaw` library ($\alpha = 2.6, x_{min} = 1.0, D = 0.058$). The likelihood-ratio test returned no statistically significant preference

Table 6: KPIs of the top-3 variants.

Variant (activities)	**Freq.**	**Median dur. (min)**
Atendimento → Triagem → Consulta → Alta	247	57
Atendimento → Triagem → Consulta → Exames de Imagem → Alta	49	101
Atendimento → Triagem → Consulta → Exames Laboratoriais → Alta	12	80

for either model ($p > 0.05$), suggesting that the tail is heavy ($\alpha = 2.6$) but does not strictly conform to a power law. Approximately 16% of distinct variants are sufficient to cover 80% of all cases, confirming that the process has a compact dominant core with a large residual of rare executions.

```
# Fit power law and compare to lognormal
fit = powerlaw.Fit(distribution, discrete=True)
R, p = fit.distribution_compare('power_law', 'lognormal')

# Power Law Fit: a = 2.604, x_min = 1.0, D = 0.058
# Power Law vs Lognormal: R = -6.854, p = 0.251
# No significant difference: both models explain the data equally well
```

A *Spearman* test confirmed a statistically significant negative association between variant length and frequency ($\rho = -0.44$, $p < 0.05$): longer variants are rarer, concentrating clinical complexity in a small subset of cases.

```
# Spearman correlation between variant length and frequency
corr, p_val = stats.spearmanr(
    variants_df["variant_length"], variants_df["frequency"])

# Spearman correlation results: r = -0.4423, p = 2.9040e-06
# Significant correlation between variant length and frequency.
```

A further χ^2 test of independence confirmed that the distribution of outlier labels across variants is not random ($p < 0.05$), indicating that process outlier status are tied to specific execution paths rather than uniformly distributed.

```
# Map cases to their variants and outlier labels
outlier_map = df.groupby("case:concept:name")["outlier_label"].first()
merged = variant_case_map.merge(outlier_map, on="case:concept:name")

merged["variant_hash"] = merged["variant"].apply(lambda x: hash(x))
ct = pd.crosstab(merged["variant_hash"], merged["outlier_label"])
chi2, p, dof, expected = stats.chi2_contingency(ct)
```

```
# Chi-squared test results: chi2=283.5929, p-value=5.0715e-19, dof=102
# Significant association between variant and outlier status.
```

Regarding reworked activity, *Medicamentos* (84 cases), *Materiais Hospitalares* (80 cases) and *Exames Laboratoriais* (35 cases) are the most frequently repeated activities, identified via:

```
rework_counts = pm4py.get_rework_cases_per_activity(df)
```

Case Size and Duration

We assume that larger cases require more time to complete. Both case size and case duration exhibit non-normal distributions (*Shapiro-Wilk*, $p < 0.05$), as visible in Figure 2. The majority of cases are short and handled quickly, while a minority of complex cases drives the upper tail.

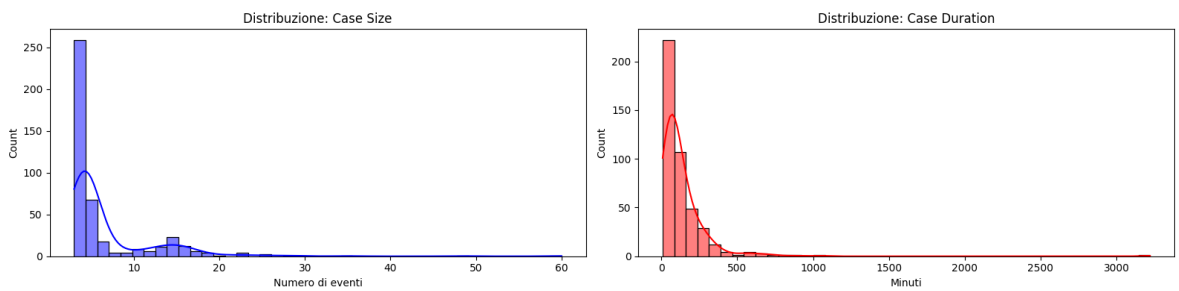


Figure 2: Distribution of case size (left) and case duration in minutes (right).

A *Spearman* correlation test between the two KPIs returned $\rho = 0.59$ ($p < 0.05$), confirming a moderate positive association: more activities reliably predict a longer stay.

```
_, p_size = stats.shapiro(kpis_all["case_size"])
_, p_duration = stats.shapiro(kpis_all["duration_min"])
corr, p_val = stats.spearmanr(
    kpis_all["case_size"], kpis_all["duration_min"])

# Case Size - Shapiro p-value: 2.2947e-32 -> Non-normal
# Case Duration - Shapiro p-value: 7.9718e-35 -> Non-normal
# Spearman correlation results: r = 0.5901, p = 2.9046e-42
# Significant correlation between case size and case duration.
```

Global Process Discovery

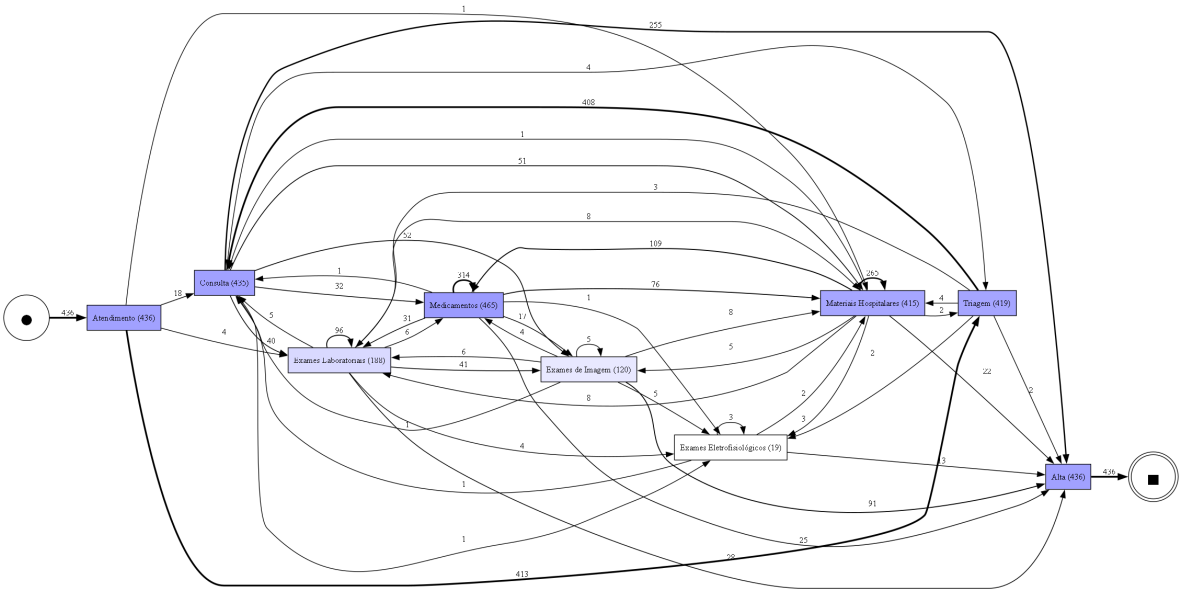


Figure 3: Global DFG annotated with event frequencies.

A Directly-Follows Graph was computed on the full log to provide an initial structural overview. The DFG reveals a hub topology centred on *Consulta* (435 occurrences), from which most clinical activities branch out. The large number of cross-edges and the self-loop on repeated activity confirm the rework patterns identified above.

The global model is necessarily confusing due to the long tail of rare variants. Subsequent segmentation analyses are therefore aimed at deriving more focused and accurate models for homogeneous sub-populations.

4.3 Outlier-Inlier Analysis

This section investigates the structural and behavioural differences between cases labelled as outliers and inliers based on the pre-computed `outlier_label` field, which encodes case duration deviation. The goal is to characterise these differences quantitatively, discover a reference process model from inliers, and use it to identify which activities drive deviant behaviour in outliers.

KPIs Analysis

KPIs (case duration, size and activity rework) were computed for each segment and compared using the *Mann-Whitney U* test, given the non-normal distributions confirmed by *Shapiro-Wilk*.

```
outliers = df[df["outlier_label"] == "outlier"]
inliers = df[df["outlier_label"] != "outlier"]

kpi_outlier = compute_case_kpis(outliers)
kpi_inlier = compute_case_kpis(inliers)

for kpi in ["duration_min", "case_size", "rework_count"]:
    _, p = stats.shapiro(kpi_outlier[kpi])
    _, p = stats.shapiro(kpi_inlier[kpi])
    res = stats.mannwhitneyu(kpi_outlier[kpi], kpi_inlier[kpi])

# Both 'duration_min' kpi are non-normal
# Significant difference between outlier and inlier for 'duration_min'
# Both 'case_size' kpi are non-normal
# Significant difference between outlier and inlier for 'case_size'
# Both 'rework_count' kpi are non-normal
# Significant difference between outlier and inlier for 'rework_count'
```

Table 7: KPI comparison between inlier and outlier segments.

	Inlier		**Outlier**	
KPI	**Mean**	**Median**	**Mean**	**Median**
Case duration (min)	100.5	78	619.0	465
Case size	6.2	4	14.8	12
Rework count	1.5	0	8.7	6

As shown also in Table 7 and Figure 4, all three KPIs differ significantly between the two segments ($p < 0.05$). The rework gap is the most informative signal: outlier cases

are not simply longer versions of normal ones, but are characterized by an accumulation of repeated diagnostic and therapeutic activities.

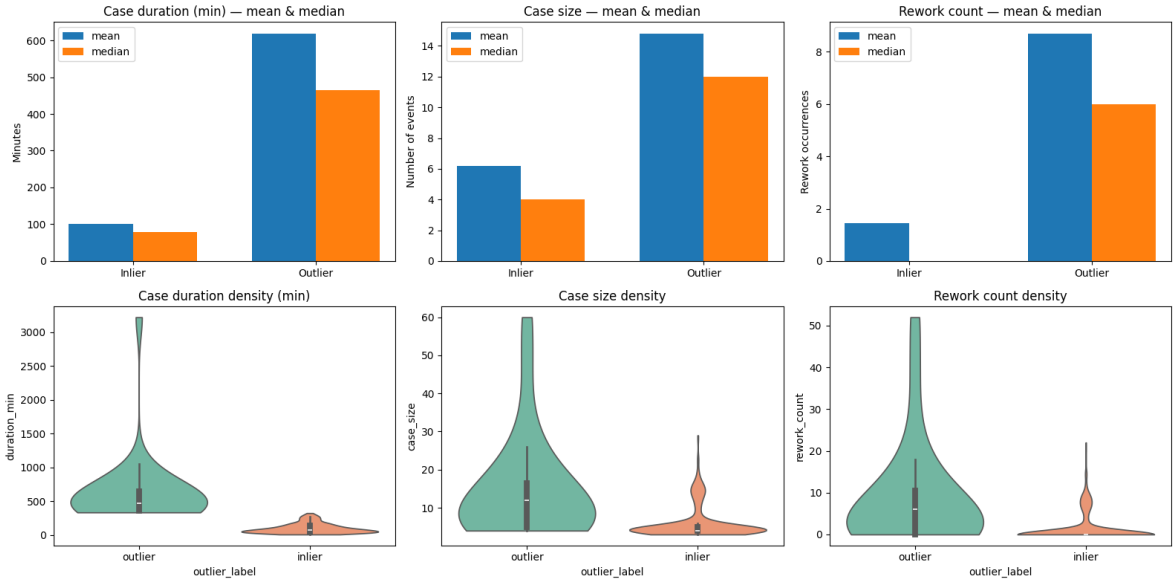


Figure 4: KPI comparison between inlier and outlier segments: case duration, case size, and rework count (mean/median and density).

Inlier Process Model

A reference model was discovered exclusively from the inlier log. To further reduce noise, only inlier variants with frequency above the power-law $x_{\min}$ cutoff (identified in Section 4.2) were retained, removing singleton traces that would otherwise inflate model complexity without adding representational value. On this filtered log, two discovery strategies were evaluated: tuning the noise threshold of the Inductive Miner, and filtering to the top-k variants. Both were optimised by maximising the F1 score (harmonic mean of fitness and precision) against the inlier log.

```
# Apply power-law cutoff to inlier log
inliers_cutoff = inliers[
    inliers["case:concept:name"].isin(cases_over_cutoff)]

# Discover and view the Petri net with the best threshold
best_threshold, best_f1 = find_best_inductive_miner_threashold(
    inliers_cutoff, lower_bound=0.0, upper_bound=0.5, step=0.1)
```

```
n_in, i_in, f_in = pm4py.discover_petri_net_inductive(inliers,
                                    best_threshold)

# Discover and view the Petri net with the best k
best_k_result = find_best_k_by_f1(
    inliers_cutoff, k_values=range(1, 11),
    discovery_algo=pm4py.discover_petri_net_inductive)
representative = pm4py.filter_variants_top_k(
    inliers_cutoff, best_k_result["best_k"])
n_in, i_in, f_in = pm4py.discover_petri_net_inductive(representative)
```

The top-k ($k = 7$) variant model achieved the best metric (fit. 0.98, prec. 0.94, f1 0.96 on inlier segment) and is shown in Figure 5. The model captures a sequential backbone (*Atendimento* → *Triagem* → *Consulta* → *Alta*) with optional parallel branches for the main diagnostic and therapeutic activities, representing the standard patient journey without the noise introduced by very rare deviant traces.

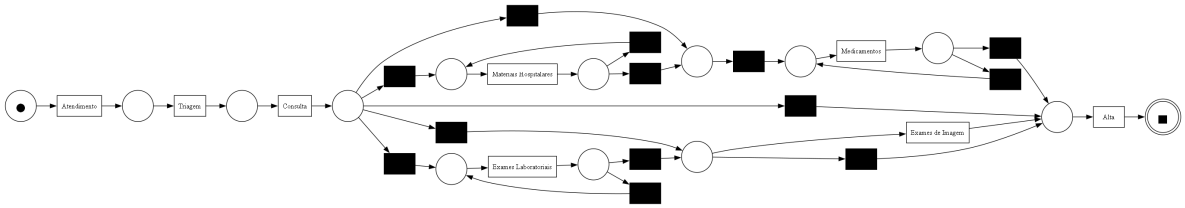


Figure 5: Petri net discovered from inlier cases (top-7 variant model, power-law cutoff applied). This model serves as the reference for all subsequent conformance checks.

When the same model is evaluated on the outlier log, metrics drop noticeably (fit. 0.94, prec. 0.79, f1 0.86 on outlier segment), confirming that outliers follow a structurally different execution pattern that the inlier model does not account for.

Conformance Check on Outliers

Alignment-based conformance checking was performed on the outlier segment against the inlier reference model. Each misalignment was classified as *insert* (activity present in the log but not expected by the model), *skip* (expected activity not executed), *early*, or *late* (activity executed out of the expected order).

```
# Conformance checking on outliers using the best inliers model
cc_out = perform_cc_al(outliers, net_inlier, im_inlier, fm_inlier)
cc_out_class = apply_classification(cc_outliers)
```

As shown in Figure 6, deviations are dominated by *insert* moves: *Exames Laboratoriais* (9 inserts) and *Exames de Imagem* (6 inserts) are the most affected, followed by *Exames Eletrofisiolgicos* (5 inserts) and *Medicamentos* (4 inserts), confirming that outlier cases execute additional diagnostic and therapeutic steps not prescribed by the standard model. *Late* moves on *Consulta* (3) and *Materiais Hospitalares* (1) indicate delayed executions. Notably, *Triagem* shows 4 *skip* moves, meaning that a share of outlier cases bypass the triage step entirely, that is a potentially critical protocol violation. Outliers therefore deviate in intensity and structure: they accumulate repeated diagnostic procedures, experience delayed consultations, and in some cases skip or delay triage altogether, pointing to clinical complexity not managed proactively and physician workload as the root causes.

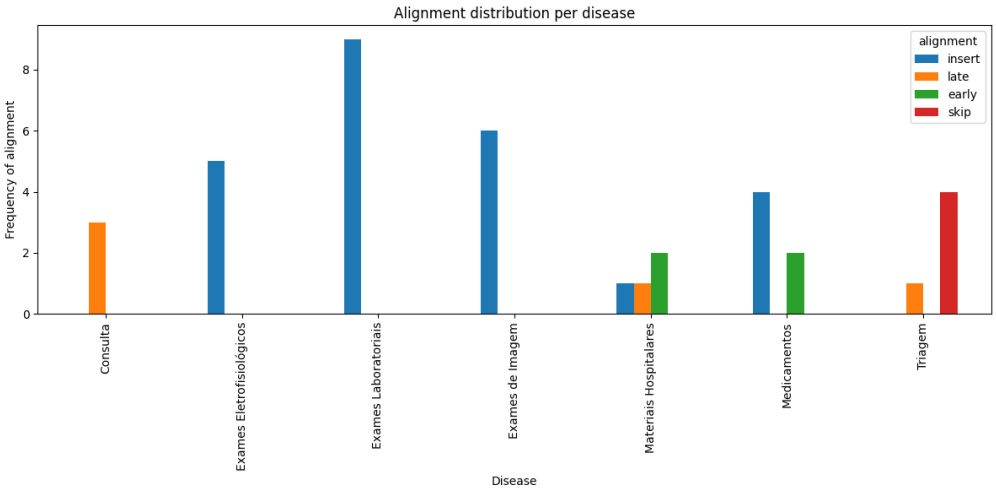


Figure 6: Alignment deviation distribution per activity in the outlier segment.

4.4 Readmission Analysis

This section investigates whether readmission is associated with a different process behaviour compared to standard discharge. The underlying assumption is that patients who return to the emergency room may have followed a less complete or less effective clinical pathway during their first visit.

KPI Analysis

KPIs were computed for each segment and compared using the *Mann-Whitney U* test, given the non-normal distributions confirmed by *Shapiro-Wilk*.

```
discharged = df[df["readmission_status"] == "Sem retorno"]
readmitted = df[df["readmission_status"] != "Sem retorno"]

kpi_discharged = compute_case_kpis(discharged)
kpi_readmitted = compute_case_kpis(readmitted)

for kpi in ["duration_min", "case_size"]:
    _, p = stats.shapiro(kpi_readmitted[kpi])
    _, p = stats.shapiro(kpi_discharged[kpi])
    res = stats.mannwhitneyu(kpi_readmitted[kpi], kpi_discharged[kpi])

# Both 'duration_min' kpi are non-normal
# Significant difference between outlier and inlier for 'duration_min'
# Both 'case_size' kpi are non-normal
# Significant difference between outlier and inlier for 'case_size'
# Both 'rework_count' kpi are non-normal
# Significant difference between outlier and inlier for 'rework_count'
```

Table 8: KPI comparison between discharged and readmitted segments.

	Discharged		**Readmitted**	
KPI	**Mean**	**Median**	**Mean**	**Median**
Case duration (min)	129.4	80	185.6	155
Case size	6.6	4	9.4	7
Rework count	1.8	0	3.8	1

As shown also in Table 8 and Figure 7, all three KPIs are significantly higher for readmitted cases ($p < 0.05$). The median duration nearly doubles (80 vs 155 minutes) and the median case size increases from 4 to 7 events. The rework gap is also present,

though less pronounced than other kpi, suggesting that readmitted patients undergo more complex treatments but without the extreme accumulation of repeated activities.

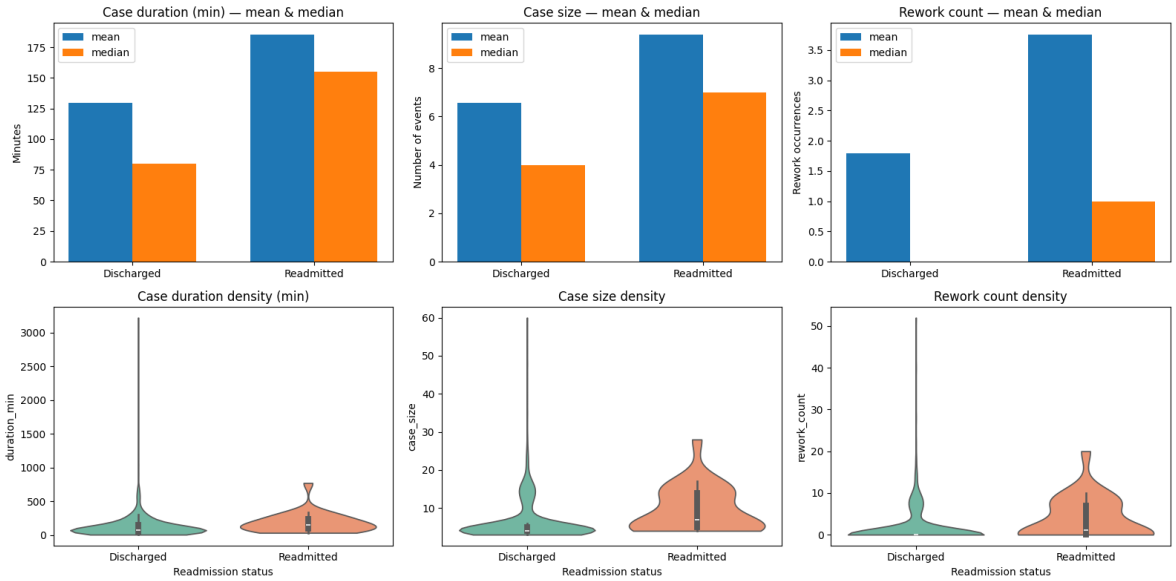


Figure 7: KPI comparison between discharged and readmitted segments: case duration, case size, and rework count (mean/median and density).

Conformance Check and Problematic Transitions

Both segments were replayed against the inlier reference model using token-based replay, and problematic transitions were extracted and compared.

Both groups show high average trace fitness on the inlier model (0.988 for discharged, 0.984 for readmitted), indicating that both follow the standard process structure. However, the percentage of fully fitting traces is lower for readmitted cases (68% vs 81%), suggesting more frequent partial deviations on complex cases. The *Wilcoxon* test on related problematic transitions sample returns a borderline result ($p > 0.05$), so no statistically significant difference can be claimed.

```
discharged_cc = perform_cc_tbr(discharged, n, i, f)
readmitted_cc = perform_cc_tbr(readmitted, n, i, f)

prob_discharged = extract_problematic_transitions(discharged_cc)
prob_readmitted = extract_problematic_transitions(readmitted_cc)

# Wilcoxon signed-rank test on related samples
```

```
stat, p = stats.wilcoxon(prob_discharged, prob_readmitted)
# Wilcoxon signed-rank test: p = 0.0625
# No difference between discharged and readmitted prob. transition.
```

As shown in Figure 8, the most frequent problematic transitions in both groups are *Materiais Hospitalares* and *Exames Laboratoriais*, consistent with the rework patterns identified in the variant and outlier analyses.

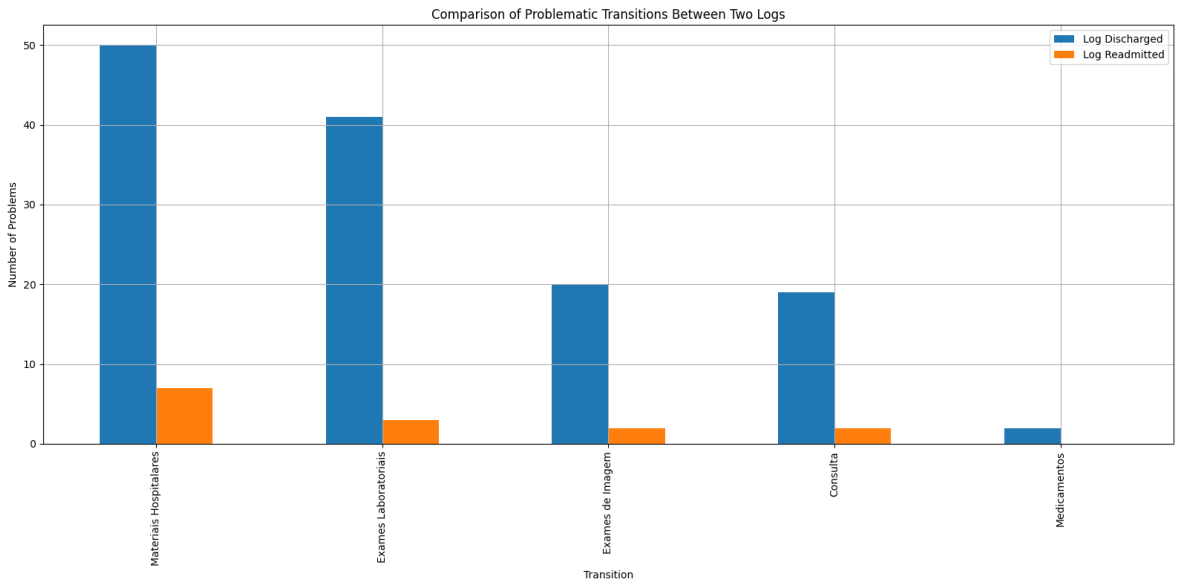


Figure 8: Problematic transitions between discharged and readmitted cases on the inlier reference model.

Overall, readmitted patients show a statistically more complex process in terms of KPIs, but their execution paths do not deviate excessively from the standard model in a way that is clearly distinguishable from discharged cases. This suggests that readmission is driven more by clinical factors than by process-level inefficiencies detectable in the event log.

4.5 Disease Analysis

This section examines whether the clinical diagnosis affects the process behaviour. The underlying assumption is that different diseases may require different diagnostic and therapeutic pathways, and that some conditions, due to their complexity or tendency to cause complications, may be associated with longer stays, more rework, and higher readmission rates.

KPI Analysis

KPIs were computed per disease and compared selecting the appropriate test after checking normality (*Shapiro-Wilk*) and homoscedasticity (*Levene*) for each group (equal std.). Since at least one group is non-normally distributed for all KPIs, the *Kruskal-Wallis* test was applied in all cases.

```
disease_kpis = df.groupby("disease").apply(compute_case_kpis)

for kpi in ["duration_min", "case_size", "rework_count"]:
    groups = [g[kpi].values for _, g
        in disease_kpis.groupby("disease")]

    all_normal = all(
        stats.shapiro(g)[1] > 0.05 for g in groups
            if len(g) >= 3)
    equal_std = stats.levene(*groups)[1] > 0.05

    if all_normal and equal_std:
        # ANOVA test
        stat, p = stats.f_oneway(*groups)
    else:
        # Kruskal-Wallis test, non-parametric alternative
        stat, p = stats.kruskal(*groups)

# Not all groups for 'duration_min' are normal or have equal var
# Significant differences between diseases for 'duration_min'
# Not all groups for 'case_size' are normal or have equal var
# Significant differences between diseases for 'case_size'
# Not all groups for 'rework_count' are normal or have equal var
# Significant differences between diseases for 'rework_count'
```

The top-3 diseases cover 97.7% of all cases: *J11.1 Influenza* (349 cases, 80%), *N23 Colica nefretica* (47 cases, 11%), and *R07.4 Dor toracica* (30 cases, 7%). Given this dominance, the *Kruskal-Wallis* test confirms significant differences across diseases for all three KPIs ($p < 0.05$).

Table 9: KPI summary per disease (top-3).

	J11.1		N23		R07.4	
KPI	**Mean**	**Median**	**Mean**	**Median**	**Mean**	**Median**
Case duration (min)	112	71	193	170	250	145.5
Case size	5	4	15	15	12.5	8
Rework count	0.6	0	8.5	8	6.2	2

Pairwise Comparison

To statistically identify which diseases drive the differences, pairwise *Mann-Whitney U* tests (assuming a non-normally distributed sample) were performed for each KPI, with *Holm* correction for multiple comparisons.

```
for kpi in kpis_to_test:
    for disease in disease_stats["disease"].unique():
        group = disease_kpis[disease_kpis["disease"] == disease][kpi]
        others = disease_kpis[disease_kpis["disease"] != disease][kpi]

        # Compare the two indipendent groups
        stat, p = stats.mannwhitneyu(group, others)
        res.append({"disease": disease, "kpi": kpi, "p_val": p})

# Adjust for multiple testing to reduce false positives chances.
res["p_adj"] = multipletests(results_df["p_val"], method="holm")

# Significant differences for 'duration_min': 3 diseases
# Significant differences for 'case_size': 3 diseases
# Significant differences for 'rework_count': 3 diseases
```

After the tests, *N23* and *R07.4* show significantly higher case duration, case size, and rework count compared to the rest ($p_{\text{adj}} < 0.05$), while *J11.1* shows significantly lower values for all three KPIs. This confirms that influenced patients follow a faster, simpler process, while renal colic and chest pain cases require substantially more diagnostic and therapeutic steps.

Readmission-Disease Correlation

The association between disease and readmission status was assessed using a global χ^2 test followed by per-disease *Fisher's exact* test with *Holm* correction.

```
corr_df = df.groupby("case:concept:name").agg(
    disease=("disease", "first"),
```

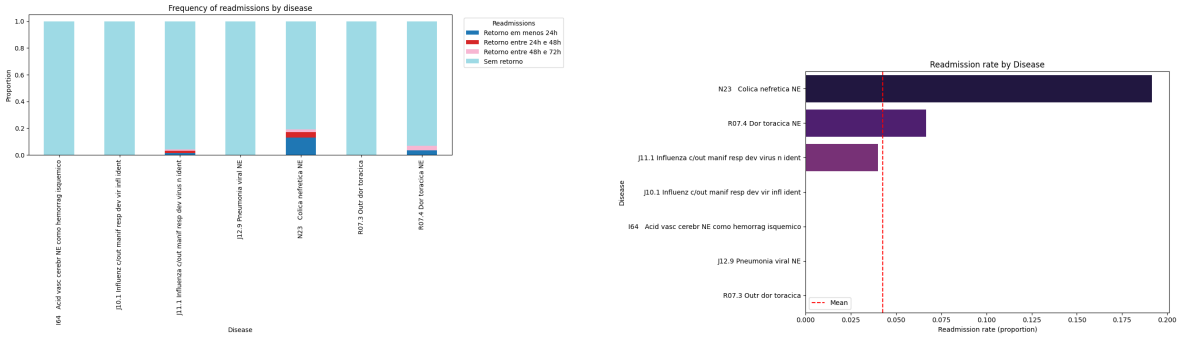
```
    readmission_status=("readmission_status", "first")
).reset_index()
ct = pd.crosstab(corr_df["disease"], corr_df["readmission_status"])
chi2, p, dof, expected = stats.chi2_contingency(ct)

for disease, group in corr_df.groupby("disease"):
    others = corr_df[corr_df["disease"] != disease]

    a = group["readmitted"].sum()    # disease & readmitted
    b = len(group) - a               # disease & not readmitted
    c = others["readmitted"].sum()   # others & readmitted
    d = len(others) - c              # others & not readmitted

    # Compare readmission rates of the specific disease vs all
    # other diseases combined (2x2 table comparison)
    odds_ratio, p_fisher = stats.fisher_exact([[a, b], [c, d]])

    res.append({
        "disease": disease,
        "odds_ratio": odds_ratio,
        "p_fisher": p_fisher
    })

res["p_adj"] = multipletests(results_df["p_fisher"], method="holm")

# Chi-squared test: chi2=27.10, p-value=0.0771, dof=18
# Some expected frequencies less than than 5, chi2 test not valid
# 2 significant associations between disease and readmission status
```

As shown in Figures 9a and 9b, *N23 Colica nefretica* stands out with an odds ratio of 5.5, patients with this diagnosis return to the emergency room 5.5 times more often than average. Conversely, *J11.1 Influenza* shows a below-average readmission rate (odds ratio 0.29), consistent with its simpler and more standardized clinical pathway.

Process Discovery and Conformance

A Petri net was discovered for each of the top-3 diseases, that are statistically different from the others, using the *Heuristics Miner*, which is more robust than the Inductive Miner on small and noisy logs.

```
diseases = {
    "J11.1": df[df["disease"] == "J11.1 Influenza"],
    "N23": df[df["disease"] == "N23   Colica nefretica NE"],
    "R07.4": df[df["disease"] == "R07.4 Dor toracica NE"]
}
```



(a) Readmission frequency by disease.

(b) Readmission rate per disease.

Figure 9: Readmission analysis by disease.

```
for disease_name, disease_log in diseases.items():
    nd, id, fd = pm4py.discover_petri_net_heuristics(disease_log)
    metrics = perform_metric(disease_log, net, im, fm)

# Fitness of each disease log on the global inlier model
for dis_name, dis_log in diseases.items():
    f = pm4py.fitness_token_based_replay(dis_log, n_in, i_in, f_in)
```

The discovered models (Figures 10, 11, 12) reflect the KPI findings. *J11.1* produces a relatively clean net with high fitness (0.97) and precision (0.88), confirming a well-defined and predictable pathway. *N23* and *R07.4* yield visibly more complex nets with multiple parallel branches after *Consulta*, and lower precision (0.59 and 0.56 respectively), reflecting the high clinical variability of these conditions.

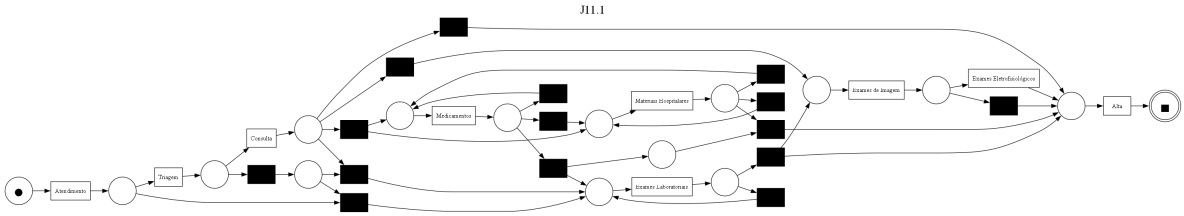


Figure 10: Petri net for *J11.1 Influenza* (Heuristics Miner).

When the top disease segments are replayed against the global inlier reference model, *J11.1* fits well (avg. fitness 0.99, 92% of traces fully fit), as expected given that it covers 80% of the cases on which the model was built. *N23* drops sharply to only 21% of fully fitting traces despite a still-high average fitness (0.95), indicating that

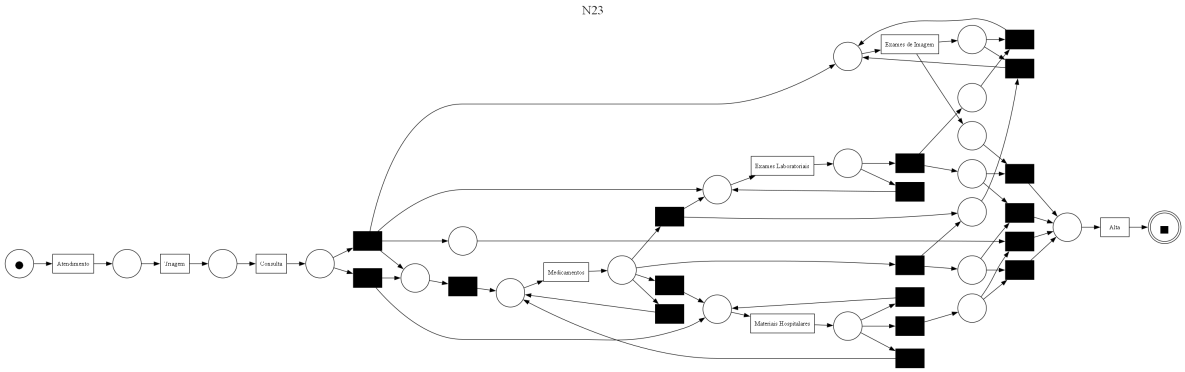


Figure 11: Petri net for *N23 Colica nefretica* (Heuristics Miner).

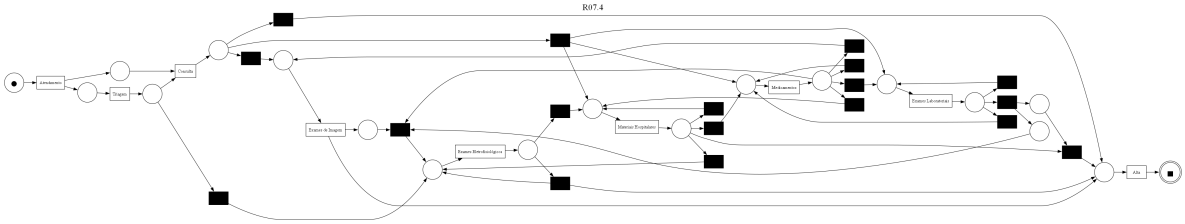


Figure 12: Petri net for *R07.4 Dor toracica* (Heuristics Miner).

while individual events are mostly replayed correctly, the overall execution sequence frequently deviates. *R07.4* sits in between (43% fully fit, avg. fitness 0.96). This confirms that the hospital's standard protocol is effectively calibrated for influenza patients and is less well-calibrated for *N23* and *R07.4* cases (also due to the smaller amount of information), which are managed under a different pathway.

4.6 Resource Analysis

In a high-volume emergency setting, it is reasonable to assume that workload is not uniformly distributed across physicians: some doctors may handle a disproportionate share of cases, which can lead to bottlenecks, delayed consultations, and ultimately longer patient stays. This section investigates whether such inequality is present and statistically significant.

To quantify this inequality, a composite workload index was defined as the weighted sum of normalised case count and total cumulative duration for each physician (with equal weights), and the *Gini* coefficient was computed over it.

```
phy_work["workload_index"] = (
    0.5 * phy_work["num_cases"] / phy_work["num_cases"].mean() +
    0.5 * phy_work["tot_duration"] / phy_work["tot_duration"].mean()
)
gini_workload = Gini(phy_work["workload_index"].values)

# Gini coefficient for physician workload distribution: 0.7367
# The workload distribution is highly unequal.
```

Figure 13 shows both the case count and the workload index per physician. The imbalance is immediately visible in both panels: *Mdico 72*, *Mdico 58*, and *Mdico 22* stand out clearly from the rest, with *Mdico 72* reaching a workload index of 10.1, more than ten times the mean ($\mu = 1.0, \sigma = 2.15$) and roughly twice that of the second most burdened physician. The *Gini* coefficient of 0.74 (> 0.5) confirms a highly unequal distribution.

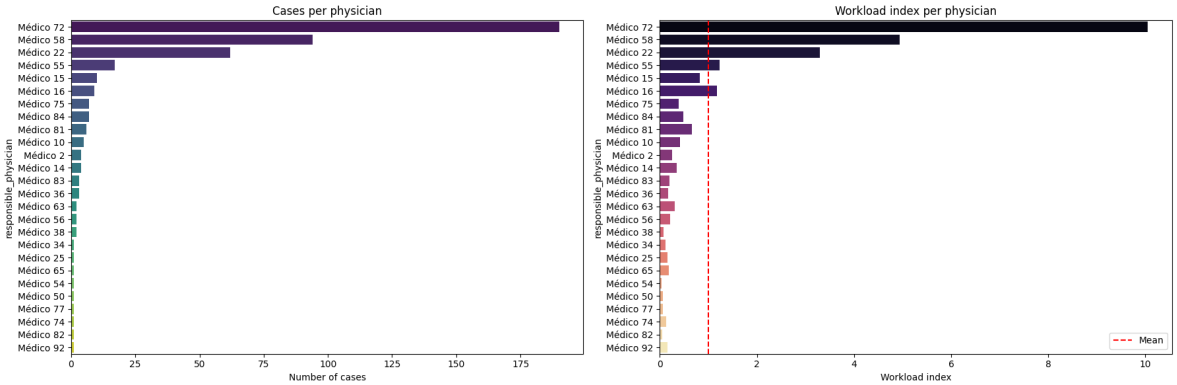


Figure 13: Cases per physician (left) and workload index per physician (right). The red dashed line marks the mean ($\mu = 1.0$) workload index.

Together, the top three physicians cover approximately 80% of all cases and 70% of

all workload, while all the remaining share the rest. This concentration is a structural risk factor: an uneven case distribution inevitably leads to delayed consultations and longer patient stays for the most burdened physicians.

5 Conclusions

The exploratory analysis at Section 4.2 revealed a highly variable process with a heavy-tailed variant distribution: a small core of simple traces coexists with a long tail of rare, complex executions. There is a significant association between variant and outlier status, this indicates that inefficiencies are not random but linked to specific execution paths of the process. We also know that the longer a variant is, the less frequent it is, so complex case are less frequent. Rework in *Medicamentos* and *Materiais Hospitalares* emerged as a recurrent structural pattern across all dataset.

The outlier-inlier analysis at Section 4.3 confirmed that deviant cases differ not only in duration but in intensity: outliers accumulate repeated diagnostic procedures, experience delayed consultations, and in some cases delay or bypass triage entirely. The inlier reference model served as a reliable benchmark for all subsequent conformance checks.

The readmission analysis at Section 4.4 showed that readmitted patients follow a statistically more complex process, but no significant structural difference in problematic transitions was detected, suggesting that readmission is driven more by clinical factors than by process-level inefficiencies.

The disease analysis at Section 4.5 confirmed that *N23 Colica nefretica* and *R07.4 Dor toracica* deviate significantly positively from the standard protocol, which is calibrated almost exclusively on *J11.1 Influenza* cases. *N23* also shows a readmission ratio higher than the average, making it the highest-risk diagnosis in the dataset, where instead *J11.1 Influenza* has a lower one, which indicates a much lower return of patients of this disease over time compared to the average. *N23* and *R07.4* also produce visibly more complex process model with lower precision, while *J11.1* yields a clean, well-defined model consistent with its dominant role in the log.

The resource analysis at Section 4.6 revealed a high *Gini* coefficient, with *Mdico 72* alone handling the majority of all cases and reaching a disproportionate workload index against the average. This extreme concentration is not only an equity issue but a direct operational risk, as overloaded physicians are structurally more likely to cause delayed consultations and longer patient stays.

Suggested improvements.

- **Enforce triage compliance:** the triage bypass observed in outlier cases is a direct protocol violation that should be addressed through system-level controls or staff training.

- **Standardise the *J11.1* pathway:** since influenza accounts for 80% of all cases,

it is reasonable to assume it is the most routinely managed condition in the emergency room. Physicians therefore already possess the clinical experience necessary to collectively define and agree on predefined care pathways covering all possible branches. Formalising this knowledge into a documented standard protocol would reduce variability and ease onboarding of new staff.

- **Define disease-specific pathways:** the current discovered process is not well-suited for the whole disease accounted, due especially to the lack of data. Dedicated and formalized clinical protocols should be defined, with the consensus of all the specialist, to reduce variability and rework.

- **Rebalance physician workload:** the extreme concentration on three physicians is a structural bottleneck. Case assignment policies should be revised to distribute load more equitably.

- **Monitor rework activities:** repeated prescriptions of *Medicamentos* and *Materiais Hospitalares* should be flagged in real time to identify clinical complexity early and intervene proactively.

- **Improve data collection:** the limited sample size different from the *J11.1 Influenza* reduces the statistical power of the disease-specific models. Collecting more granular and longitudinal data would enable more robust analysis and predictive modelling.

References

[1] Luca Milanesi. *GitHub project repository*. 2026. URL: https://github.com/Luca-02/brazilian-hospital-process-mining.